Perimeter & Area

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: A rectangle is 6 cm long and 4 cm wide. What is its perimeter?

Answer: Add all 4 sides: 6 + 4 + 6 + 4 = 20 cm.

Standard: A rectangle is 5 cm long and 3 cm wide. What is its area?

Answer: Area = length × width = 5 × 3 = 15 cm².

Advanced: A square has a perimeter of 20 cm. What is the length of one side?

Answer: A square has 4 equal sides, so 20 ÷ 4 = 5 cm.

Easy / Standard (15 questions)

1. A rectangle is 5 cm by 3 cm. What is its perimeter?
2. A square has sides of 4 cm. What is its perimeter?
3. A rectangle is 6 cm by 2 cm. What is its area?
4. Which shape has a perimeter of 20cm? A) a 6cm×4cm rectangle B) a square of side 5cm C) both A and B D) neither
5. A rectangle is 8 cm long and 2 cm wide. What is its perimeter?
6. A square has a perimeter of 12 cm. What is the length of one side?
7. What is the area of a 4cm by 4cm square? A) 8cm² B) 12cm² C) 16cm² D) 20cm²
8. A square has an area of 9 cm². What is the length of one side?
9. Select ALL of these shapes that have a perimeter LESS than 18cm: A) a 4cm×3cm rectangle B) a square of side 5cm C) a 3cm×2cm rectangle D) a 6cm×5cm rectangle E) a square of side 4cm
10. A rectangle has a length of 7 cm and a width of 3 cm. What is its perimeter?
11. A rectangle is 9 cm long and 1 cm wide. What is its area?
12. Which rectangle has the larger area? A) 5cm×3cm B) 4cm×4cm C) they are equal D) cannot tell
13. A square has sides of 6 cm. What is its perimeter?
14. A rectangle is 5 cm by 5 cm. What shape is it, and what is its area?
15. A rectangle has an area of 12 cm² and a width of 3 cm. What is its length?

Standard (15 questions)

1. A rectangle is 7 cm by 4 cm. What is its perimeter?
2. Which rectangle has a perimeter of 22cm? A) 5cm×6cm B) 8cm×3cm C) both A and B D) neither
3. A rectangle is 6 cm by 5 cm. What is its area?

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4. A square has an area of 25 cm^2. What is its perimeter?

5. A rectangle has a perimeter of 26 cm and a length of 8 cm. What is its width?

6. What is the area of a rectangle 9cm long and 3cm wide? A) 24cm^2 B) 27cm^2 C) 30cm^2 D) 12cm^2

7. A square garden has sides of 12 m. What is its perimeter?

8. Select ALL of these shapes that have an area of MORE than 20cm^2: A) 6cm×4cm B) 5cm×5cm C) 3cm×6cm D) 7cm×3cm E) 8cm×2cm

9. A rectangle is 10 cm long and 4 cm wide. What is its area?

10. Which shape has the SAME area as a 6cm×4cm rectangle? A) an 8cm×3cm rectangle B) a 24cm^2 square (side √24) C) both A and B D) neither

11. A rectangle has a width of 5 cm and its length is double the width. What is its perimeter?

12. A square has a perimeter of 36 cm. What is its area?

13. Two rectangles, 4cm×3cm and 2cm×3cm, are joined edge to edge with no overlap along their 3cm sides. What is the total area?

14. A rectangular field is 30 m by 15 m. What is its perimeter?

15. A rectangle has an area of 32 cm^2 and a length of 8 cm. What is its width?

Standard / Hard (15 questions)

1. A gardener has a 10 m by 6 m rectangular lawn. He cuts a 2m×2m patio out of one corner of it. How much lawn area is left?

2. Working out which shape has an area of exactly 30cm^2, pick the correct one. A) a 6cm×6cm square with a 2cm×3cm piece removed B) a 5cm×6cm rectangle C) a 3cm×7cm rectangle D) a 24cm^2 triangle

3. Pushed right up against each other, two square tiles with sides of 5 cm form a single rectangle. What perimeter does that rectangle have?

4. A rectangle's width is 4 cm, and its length is 3 cm longer than that. Work out the rectangle's perimeter.

5. Carpeting costs £8 for every square metre. How much would it cost altogether to carpet a room measuring 5 m by 4 m?

6. The area inside a square garden comes to 36 m^2. What distance is it all the way around the garden's edge?

7. Looking at a 10cm×5cm rectangle, select ALL of the statements below that are TRUE about it: A) its perimeter is 30cm B) its area is 50cm^2 C) its perimeter is 50cm D) it has 4 right angles E) its area is 30cm^2

8. Starting with a 20cm×10cm piece of card, a 4cm×3cm rectangle is cut away from just one of its corners. What area of card is left over?

9. An L-shape is formed when a 4m×2m rectangle is removed from the corner of a bigger 12m×5m rectangle. Work out the area of the L-shape that remains.

10. Measuring 50 m long and 30 m wide, a sports pitch is rectangular in shape. What is the distance all the way around its edge? A) 80 m B) 140 m C) 160 m D) 150 m

11. A rectangle and a square cover the exact same area of 36 cm^2. Given that the rectangle's width is 4 cm, what must its length be?

12. Out of the four rectangles (or square) described below, which one has the biggest perimeter? A) a 6cm×6cm square B) a 4cm×8cm rectangle C) a 5cm×9cm rectangle D) a 3cm×10cm rectangle

13. A rectangular room is made of two parts: a 4m×3m rectangle and a 2m×2m square, with no overlap. What is the total area of the room?

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14. A rectangular banner is 5cm by 3cm. If both its length and width are doubled, what is the new area?

15. A square field is 10m by 10m. A path 1m wide is built all the way around the OUTSIDE of the field. What is the perimeter of the path's outer edge?

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